

Generation Interconnection System Impact Study Report

For Queue Project Q21-1672

Lockwood Rd 100 kV

February 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1672. The Interconnection Customer proposes a 185.5 MW solar photovoltaic facility interconnecting at the Lockwood Rd 100 kV substation in the Ohio area. The request was received on February 16, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 610 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$95,956,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

LAKE SERENE - LOCKWOOD RD 100.0 kV: 142.1% loading (124 MVA) following the outage of the LOCKWOOD RD - MAUCKPORT 100.0 kV; DFAX = 100.0%.

SPEARMAN CITY - WELTON MOHAWK 100.0 kV: 141.1% loading (110 MVA) following the outage of the SPEARMAN CITY - SYLMAR E 100.0 kV Line; DFAX = 6.6%.

LAKE SERENE - MOSAIC CO TAMPA FACILITY 100.0 kV: 140.1% loading (121 MVA) following the outage of the LOCKWOOD RD - MAUCKPORT 100.0 kV Line; DFAX = 100.0%.

The CITY OF BRYAN - MOSAIC CO TAMPA FACILITY 100.0 kV Line is overloaded to 139.4% of its 116 MVA emergency rating for the loss of the LOCKWOOD RD - MAUCKPORT 100.0 kV. The project contributes 100.0% DFAX (185.5 MW) to this overload.

The SPEARMAN CITY - SYLMAR E SUBSTATION 100.0 kV line is overloaded to 138.3% of its 122 MVA emergency rating for the loss of the SPEARMAN CITY - WELTON MOHAWK 100.0 kV. The project contributes 7.2% DFAX (13.3 MW) to this overload.

MAUCKPORT - LOCKWOOD RD 100.0 kV: 135.8% loading (130 MVA) following the outage of the LOCKWOOD RD - LAKE SERENE 100.0 kV L; DFAX = 100.0%.

For the loss of SWA VLY SUBSTATION - MUSKOGEE SUBSTATION 500.0 kV, the DUTTON PUMP - CELINA SOLAR PROJECT 1 LLC 100.0 kV loads to 134.3% of its 133 MVA rating. Project DFAX: 7.7%.

CITY OF BRYAN - MAYBERRY 100.0 kV: 132.9% loading (109 MVA) following the outage of the LOCKWOOD RD - MAUCKPORT 100.0 kV line; DFAX = 100.0%.

For the loss of SPEARMAN CITY - WELTON MOHAWK SUBSTATION 100.0 kV, the SYLMAR E - CELINA SOLAR PROJECT 1 LLC 100.0 kV loads to 130.4% of its 125 MVA rating. Project DFAX: 7.2%.

The BORDER TAP SUBSTATION - HESPERIA 100.0 kV line is overloaded to 130.0% of its 105 MVA emergency rating for the loss of the LOCKWOOD RD - MAUCKPORT 100.0 kV. The project contributes 100.0% DFAX (185.5 MW) to this overload.

The BORDER TAP - MAYBERRY 100.0 kV is overloaded to 128.4% of its 113 MVA emergency rating for the loss of the MAUCKPORT - LOCKWOOD RD 100.0 kV. The project contributes 100.0% DFAX (185.5 MW) to this overload.

For the loss of LAKE SERENE - LOCKWOOD RD 100.0 kV, the MAUCKPORT - CHIMACUM 100.0 kV Line loads to 120.6% of its 134 MVA rating. Project DFAX: 100.0%.

The CELINA SOLAR PROJECT 1 LLC - WELTON MOHAWK 100.0 kV line is overloaded to 118.0% of its 119 MVA emergency rating for the loss of the SYLMAR EAST - SPEARMAN CITY 100.0 kV Line. The project contributes 6.6% DFAX (12.2 MW) to this overload.

The LAKESIDE 2 - COVANTA HENNEPIN ENERGY SUBSTATION 100.0 kV Line is overloaded to 108.2% of its 175 MVA emergency rating for the loss of the SWAN VLY SUBSTATION - TREGO 500.0 kV. The project contributes 22.1% DFAX (41.0 MW) to this overload.

COVANTA HENNEPIN ENERGY SUBSTATION - CHUB LAKE 100.0 kV line: 106.8% loading (190 MVA) following the outage of the TREGO - SWAN VLY 500.0 kV L; DFAX = 22.1%.

HESPERIA - CHUB LAE 100.0 kV L: 105.3% loading (210 MVA) following the outage of the TREGO - SWAN VALLEY 500.0 kV Line; DFAX = 22.1%.

For the loss of EAST DRAINAGE CANAL - ST. LAWRENCE 230.0 kV, the BEAVERLEY - GEISEL MONUMENT 100.0 kV L loads to 101.1% of its 187 MVA rating. Project DFAX: 5.1%.

SOUTHWEST LIMA - BEAVERLEY 100.0 kV L: 100.9% loading (199 MVA) following the outage of the E DRAINAGE CAAL - ST. LAWRENCE 230.0 kV line; DFAX = 5.1%.

LENKURT - SOUTHWEST LIMA 100.0 kV: 100.7% loading (210 MVA) following the outage of the E DRAINAGE CANAL - ST. LAWRENCE 230.0 kV; DFAX = 5.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAKE SERENE - LOCKWOOD RD 100.0 kV L	Rebuild line with higher-capacity conductor	\$8,533,000	100% (\$8,533,000)
SPEARMAN CITY - WELTON MOHAWK 100.0 kV line	Rebuild line with higher-capacity conductor	\$4,621,000	100% (\$4,621,000)
LAKE SERENE - MOSAIC CO TAMPA FACILITY 100.0 kV	Rebuild line with higher-capacity conductor	\$5,797,000	100% (\$5,797,000)
CITY OF BRYAN - MOSAIC CO TAMPA FACILITY 100.0 kV	Rebuild line with higher-capacity conductor	\$5,933,000	100% (\$5,933,000)
SPEARMAN CITY - SYLMAR E 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,499,000	100% (\$2,499,000)
MAUCKPORT - LOCKWOOD RD 100.0 kV	Rebuild line with higher-capacity conductor	\$9,619,000	100% (\$9,619,000)
DUTTON PUMP - CELINA SOLAR PROJECT 1 LLC 100.0 kV	Rebuild line with higher-capacity conductor	\$5,039,000	100% (\$5,039,000)
MAYBERRY - CITY OF BRYAN 100.0 kV	Rebuild line with higher-capacity conductor	\$364,000	100% (\$364,000)
SYLMAR E - CELINA SOLAR PROJECT 1 LLC 100.0 kV	Rebuild line with higher-capacity conductor	\$6,756,000	100% (\$6,756,000)
BORDER TAP SUBSTATION - HESPERIA 100.0 kV line	Rebuild line with higher-capacity conductor	\$11,420,000	100% (\$11,420,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MAYBERRY - BRDER TAP 100.0 kV	Rebuild line with higher-capacity conductor	\$15,595,000	100% (\$15,595,000)
CHIMACUM - MAUCKPORT 100.0 kV Line	Rebuild line with higher-capacity conductor	\$7,517,000	100% (\$7,517,000)
CELINA SOLAR PROJECT 1 LLC - WELTON MOHAWK 100.0 kV line	Rebuild line with higher-capacity conductor	\$6,564,000	100% (\$6,564,000)
LAKE SIDE 2 - COVANTA HENNEPIN ENERGY 100.0 kV line	Reconductor line	\$2,202,000	46% (\$1,018,000)
COVANTA HENNEPIN ENERGY SUBSTATION - CHUB LAKE 100.0 kV line	Reconductor line	\$1,832,000	46% (\$847,000)
HESPERIA SUBSTATION - CHUB LAKE 100.0 kV L	Reconductor line	\$1,282,000	46% (\$593,000)
BEAVERLEY - GEISEL MONUMENT 100.0 kV	Reconductor line	\$825,000	100% (\$825,000)
BEAVERLEY SUBSTATION - SOUTHWEST LIMA 100.0 kV	Reconductor line	\$808,000	100% (\$808,000)
LENKURT - SOUTHWEST LIMA 100.0 kV	Reconductor line	\$1,608,000	100% (\$1,608,000)
	Total allocated to project		\$95,956,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.